

In the article "Muon Optimizer Guide: Quick Start and Key Details," we listed several versions of Muon, differing mainly in the scaling factor related to the shape of the learning rate matrix. The "official version (Keller Jordan version)" only differs from the "MuP version" by having an additional truncation operation of $\max(1, \cdot)$. This article will specifically discuss the origin of this truncation operation. $\max(1, \cdot)$ The truncation operation. This article will specifically discuss how this truncation operation came about.

Several versions

Muon's update rules can be uniformly written as

$$\begin{aligned} \mathbf{M}_t &= \beta \mathbf{M}_{t-1} + \mathbf{G}_t \\ \mathbf{W}_t &= \mathbf{W}_{t-1} - \eta_t (\alpha \operatorname{msign})(\mathbf{M}_t) + \lambda \mathbf{W}_{t-1} \end{aligned}$$

The difference between several different versions lies in α , which are: $\alpha = \left\{ \begin{array}{l} 1 \end{array} \right.$

$$\begin{aligned} \mathbf{M}_t &= \beta \mathbf{M}_{t-1} + \mathbf{G}_t \\ \mathbf{W}_t &= \mathbf{W}_{t-1} - \eta_t (\alpha \operatorname{msign})(\mathbf{M}_t) + \lambda \mathbf{W}_{t-1} \end{aligned} \tag{1}$$

α They are:

$$\alpha = \begin{cases} 1 & \text{(Simple version)} \\ \sqrt{\max(1, d_{ou}/d_{in})} & \text{(KellerJordan version)} \\ \sqrt{d_{ou}/d_{in}} & \text{(MuP version)} \\ 0.2 \times \sqrt{\max(d_{ou}, d_{in})} & \text{(Moonlight version)} \end{cases}$$

Where the matrix $\mathbf{W} \in \mathbb{R}^{d_{in} \times d_{out}}$ represents the training parameters of the linear layer $\mathbf{y} = \mathbf{x} \mathbf{W}$, and the input $\mathbf{x} \in \mathbb{R}^{d_{in}}$ is a row vector. $\mathbf{W} \in \mathbb{R}^{d_{in} \times d_{out}}$, representing a linear layer $\mathbf{y} = \mathbf{x} \mathbf{W}$. The training parameters, where the input $\mathbf{x} \in \mathbb{R}^{d_{in}}$ is a row vector.

This article primarily focuses on the "KellerJordan version" and the "MuP version," the former adding an extra $\max(1, \cdot)$ to the latter. Based on our analysis in articles such as "Higher-Order MuP: A Simpler Yet More Elegant Spectral Conditional Scaling" and "Beyond MuP: 2. Linear Layers and Steepest Descent," under MuP-related spectral conditional constraints, steepest descent should be the MuP version of Muon. How then should we interpret the extra $\max(1, \cdot)$? $\max(1, \cdot)$ According to our $\max(1, \cdot)$ How can this be explained?

Feature Increment

For simplicity, the subscript t will be omitted in the following discussion. Without loss of generality, we assume that the momentum $\mathbf{M}$ is full rank, then the singular values of $\Phi = \operatorname{msign}(\mathbf{b}$

oldsymbol{M})\$ are all 1. Therefore, when $d_{in} \leq d_{out}$, $\Phi \Phi^T = I_{d_{in}}$, and when $d_{in} > d_{out}$, $\Phi \Phi^T = I_{d_{out}}$. Without loss of generality, we assume momentum M is full rank, then $\Phi = \text{msign}(M)$ all singular values are 1, then when $d_{in} \leq d_{out}$, $\Phi \Phi^T = I_{d_{in}}$, when $d_{in} > d_{out}$, $\Phi^T \Phi = I_{d_{out}}$.

Let $\Delta W = \eta \alpha \Phi$. What we need to do is find the relationship between α and d_{in}, d_{out} . From "Why Do We Prefer Isotropy? Based on the Understanding of Steepest Descent", we know that parameters are actually just a byproduct of the model; changes at the feature level may be more fundamental. Transforming ΔW to the feature level gives $\Delta y = x \Delta W = \eta \alpha x \Phi$, then $\|\Delta y\|_{RMS} = \alpha \|\Phi x\|_{RMS}$. $\Delta W = \eta \alpha \Phi$ What we need to do is find α and d_{in}, d_{out} . The relationship. From ΔW Transforming to the feature level is $\Delta y = x \Delta W = \eta \alpha x \Phi$, So $\|\Delta y\|_{RMS} = \alpha \|\Phi x\|_{RMS}$.

Next, we need to discuss different cases. First, when $d_{in} \leq d_{out}$, Φ can be written in the form $U \begin{bmatrix} I_{d_{in}} & 0 \end{bmatrix} V^T$, where $U \in \mathbb{R}^{d_{in} \times d_{in}}$ and $V \in \mathbb{R}^{d_{out} \times d_{out}}$ are both orthogonal matrices. Then, Note that the entire expression uses equal signs, so we only need to set $\alpha = \sqrt{d_{out}/d_{in}}$ to make "each" Δy The RMS of y is $\eta \alpha \|\Phi x\|_{RMS}$, meaning that the relative update magnitude of all tokens is consistent. $d_{in} \leq d_{out}$, Φ It can be written as $U \begin{bmatrix} I_{d_{in}} & 0 \end{bmatrix} V^T$ In the form of, $U \in \mathbb{R}^{d_{in} \times d_{in}}, V \in \mathbb{R}^{d_{out} \times d_{out}}$ Since they are all orthogonal matrices, then

$$\begin{aligned} \|\Delta y\|_{RMS} &= \eta \alpha \|\Phi x\|_{RMS} = \eta \alpha \|U \begin{bmatrix} I_{d_{in}} & 0 \end{bmatrix} V^T x\|_{RMS} \\ &= \eta \alpha \|U \begin{bmatrix} I_{d_{in}} & 0 \end{bmatrix} x\|_{RMS} \\ &= \eta \alpha \|\begin{bmatrix} x & 0 \end{bmatrix}\|_{RMS} \\ &= \eta \alpha \sqrt{\frac{d_{in}}{d_{out}}} \|x\|_{RMS} \\ &= \eta \alpha \sqrt{\frac{d_{in}}{d_{out}}} \|x\|_{RMS} \end{aligned} \tag{2}$$

$\alpha = \sqrt{d_{out}/d_{in}}$ That way, "everyone" can be made Δy RMS are all $\eta \|x\|_{RMS}$. This means that the relative update magnitude of all tokens is consistent.

Isotropic

Unfortunately, for the second case $d_{in} > d_{out}$, the goal of "perfect consistency" cannot be achieved. Specifically, the SVD of Φ should be written as $U \begin{bmatrix} I_{d_{out}} & 0 \end{bmatrix} V^T$, so

in{equation}\begin{aligned} \mathbf{x} \mathbf{U} is a d_{in} -dimensional vector, $d_{in} > d_{out}$, so $(\mathbf{x} \mathbf{U})_{[:d_{out}]}$ only takes the first d_{out} dimensions of $\mathbf{x} \mathbf{U}$ to calculate the RMS. At this time, its RMS is uncertain. The maximum can be $\sqrt{d_{in}/d_{out}} \|\mathbf{x}\|_{RMS}$ (Worst Case), and the minimum can be 0. Therefore, the aforementioned goal of "complete consistency" cannot be achieved. Specifically, at this point...

The SVD should be written as $\mathbf{U} \begin{bmatrix} \mathbf{I}_{d_{out}} \\ \mathbf{0}_{(d_{in}-d_{out}) \times d_{out}} \end{bmatrix} \mathbf{V}^\top$, then

$$\begin{aligned} \|\Delta \mathbf{y}\|_{RMS} &= \eta \alpha \left\| \mathbf{x} \mathbf{U} \begin{bmatrix} \mathbf{I}_{d_{out}} \\ \mathbf{0}_{(d_{in}-d_{out}) \times d_{out}} \end{bmatrix} \mathbf{V}^\top \right\|_{RMS} \\ &= \eta \alpha \left\| \mathbf{x} \mathbf{U} \begin{bmatrix} \mathbf{I}_{d_{out}} \\ \mathbf{0}_{(d_{in}-d_{out}) \times d_{out}} \end{bmatrix} \right\|_{RMS} \\ &= \eta \alpha \left\| (\mathbf{x} \mathbf{U})_{[:d_{out}]} \right\|_{RMS} \end{aligned} \quad (3)$$

It is d_{in} dimensional vector, $d_{in} > d_{out}$, so $(\mathbf{x} \mathbf{U})_{[:d_{out}]}$ It was just a cut-out. The former d_{out} When calculating the RMS using multiple dimensions, the RMS value is uncertain, and can potentially reach a maximum of $\sqrt{d_{in}/d_{out}} \|\mathbf{x}\|_{RMS}$ (Worst Case), the smallest possible value is 0.

We know that orthogonal matrices do not change the RMS, so $\|\mathbf{x} \mathbf{U}\|_{RMS} = \|\mathbf{x}\|_{RMS}$. When the distribution of $\mathbf{x}$ is sufficiently isotropic, we can assume that this means the average scale of each component of $\mathbf{x} \mathbf{U}$ is $\|\mathbf{x}\|_{RMS}$. Therefore, taking the first d_{out} components for RMS, on average, it is also approximately equal to $\|\mathbf{x}\|_{RMS}$, that is, $\|\Delta \mathbf{y}\|_{RMS} \approx \alpha \|\mathbf{x}\|_{RMS}$. Therefore, we only need to take $\alpha = 1$ to achieve a similar effect as in the previous section. When the distribution is sufficiently isotropic, we can consider this to represent $\mathbf{x} \mathbf{U}$. The average scale of each component is $\|\mathbf{x}\|_{RMS}$. So they took the front d_{out} . Each unit, when converted to RMS, is approximately equal to $\|\mathbf{x}\|_{RMS}$. Right now $\|\Delta \mathbf{y}\|_{RMS} \approx \alpha \|\mathbf{x}\|_{RMS}$. Therefore, only need to take $\alpha = 1$. This will achieve a similar effect to the previous section.

Anisotropy

Combining the results from the first two sections, we get

$\alpha = \sqrt{\max(1, \frac{d_{out}}{d_{in}})}$, which is exactly the $\max(1, \dots)$ that appears in KellerJordan's version of Muon.

$$\alpha = \sqrt{\max \left(1, \frac{d_{out}}{d_{in}} \right)} \quad (4)$$

$\max(1, \dots)$

However, the conclusion of the previous section relies on the assumption that the input $\mathbf{x}$ is sufficiently isotropic. This may be approximately true in the early stages of training, but as training progresses, the dis

tribution of features gradually becomes anisotropic, concentrating on the "Worst Case" that maximizes $\|\Delta y\|_{\text{RMS}} / \|\Delta x\|_{\text{RMS}}$. At this point, the average approximation $\|\Delta y\|_{\text{RMS}} / \|\Delta x\|_{\text{RMS}} \approx \eta \alpha \|\Delta y\|_{\text{RMS}} / \|\Delta x\|_{\text{RMS}}$ is no longer accurate, and the maximum value $\eta \alpha \sqrt{d_{\text{in}} / d_{\text{out}}} \|\Delta y\|_{\text{RMS}} / \|\Delta x\|_{\text{RMS}}$ will be more accurate. The assumption of sufficient isotropy may hold approximately in the early stages of training, but as training progresses, the distribution of features will gradually become anisotropic, concentrating on... $\|\Delta y\|_{\text{RMS}} \approx \eta \alpha \|\Delta y\|_{\text{RMS}}$ On the maximized "Worst Case", the average approximation is as follows. $\|\Delta y\|_{\text{RMS}} \approx \eta \alpha \|\Delta y\|_{\text{RMS}}$ That's not accurate enough; it's actually the maximum value. $\eta \alpha \sqrt{d_{\text{in}} / d_{\text{out}}} \|\Delta y\|_{\text{RMS}}$ It will be more accurate.

In this case, let α of $\|\Delta y\|_{\text{RMS}} / \|\Delta x\|_{\text{RMS}} \approx \eta \alpha \|\Delta y\|_{\text{RMS}} / \|\Delta x\|_{\text{RMS}}$ be $\sqrt{d_{\text{out}} / d_{\text{in}}}$, which is consistent with the conclusion when $d_{\text{in}} \leq d_{\text{out}}$, thus obtaining the MuP version result again. In other words, for the mid-to-late stages of training, the MuP version of Muon is more scientific. To address this inconsistency, we have two strategies: First, we consistently use the MuP version of Muon, which slightly reduces the initial convergence speed, but the mid-to-late stages are crucial for training. Second, we change the scaling factor to $\alpha = \sqrt{\max(\tau_t, \frac{d_{\text{out}}}{d_{\text{in}}})}$, where τ_t monotonically decays from 1 to 0. This achieves a gradual transition from the KellerJordan version to the MuP version, at the cost of adding a Schedule to adjust.

$\|\Delta y\|_{\text{RMS}} \approx \eta \|\Delta y\|_{\text{RMS}}$ of $\eta \alpha \sqrt{d_{\text{out}} / d_{\text{in}}}$ and $d_{\text{in}} \leq d_{\text{out}}$. The conclusions were consistent, leading to the revised MuP version results. In other words, the MuP version of Muon is more scientifically sound for the mid-to-late stages of training. To address this inconsistency, we have two strategies: first, consistently use the MuP version of Muon, which will slightly reduce the initial convergence speed, but the mid-to-late stages are crucial for training; second, change the scaling factor to...

$$\alpha = \sqrt{\max\left(\tau_t, \frac{d_{\text{out}}}{d_{\text{in}}}\right)} \quad (5)$$

τ_t The transition from 1 to 0 is monotonically decayed, thus achieving a gradual change from the Keller Jordan version to the MuP version. The trade-off is that an additional schedule needs to be adjusted.

Article Summary

This article mainly explains the origin of $\max(1, \cdot)$ in the KellerJordan version from the perspective of the uniformity of "feature increment". $\max(1, \cdot)$ The source of.

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